

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

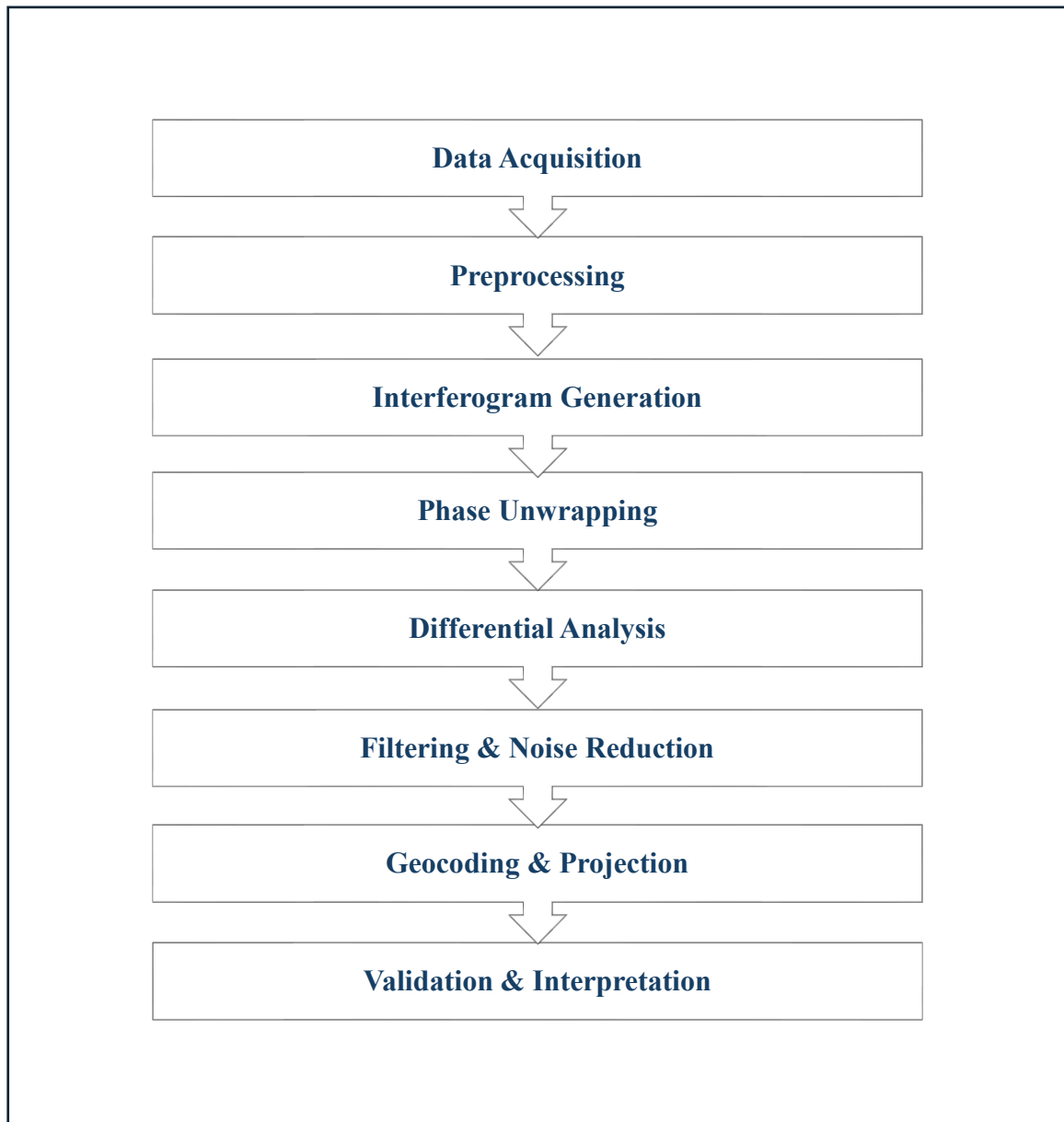
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
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S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 1006+100

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

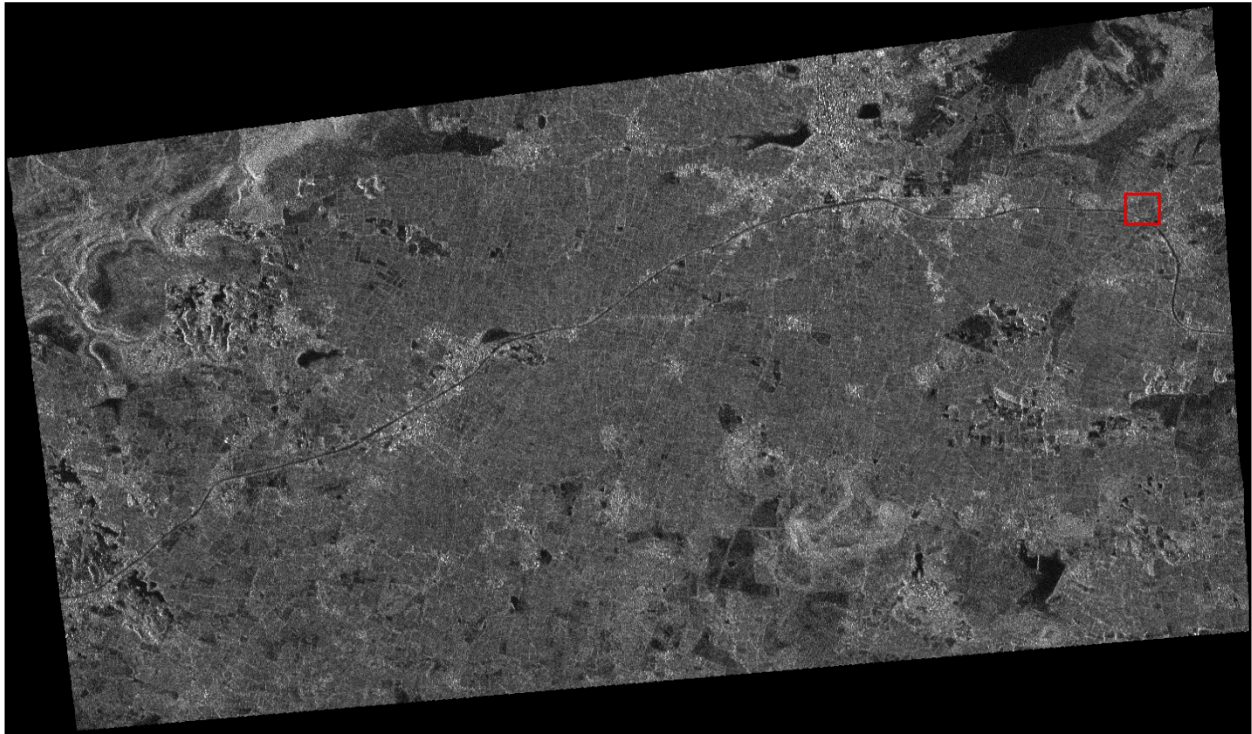
Flight Direction - DESCENDING

Polarization – VV+VH

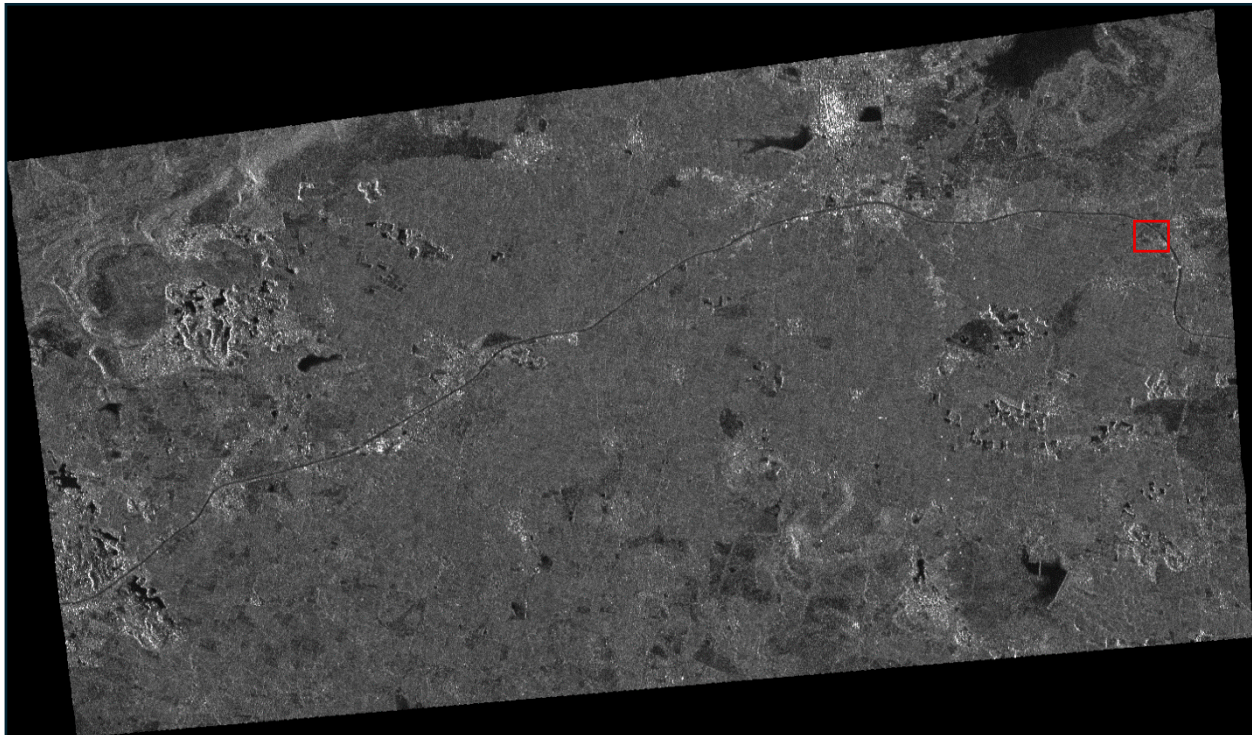
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

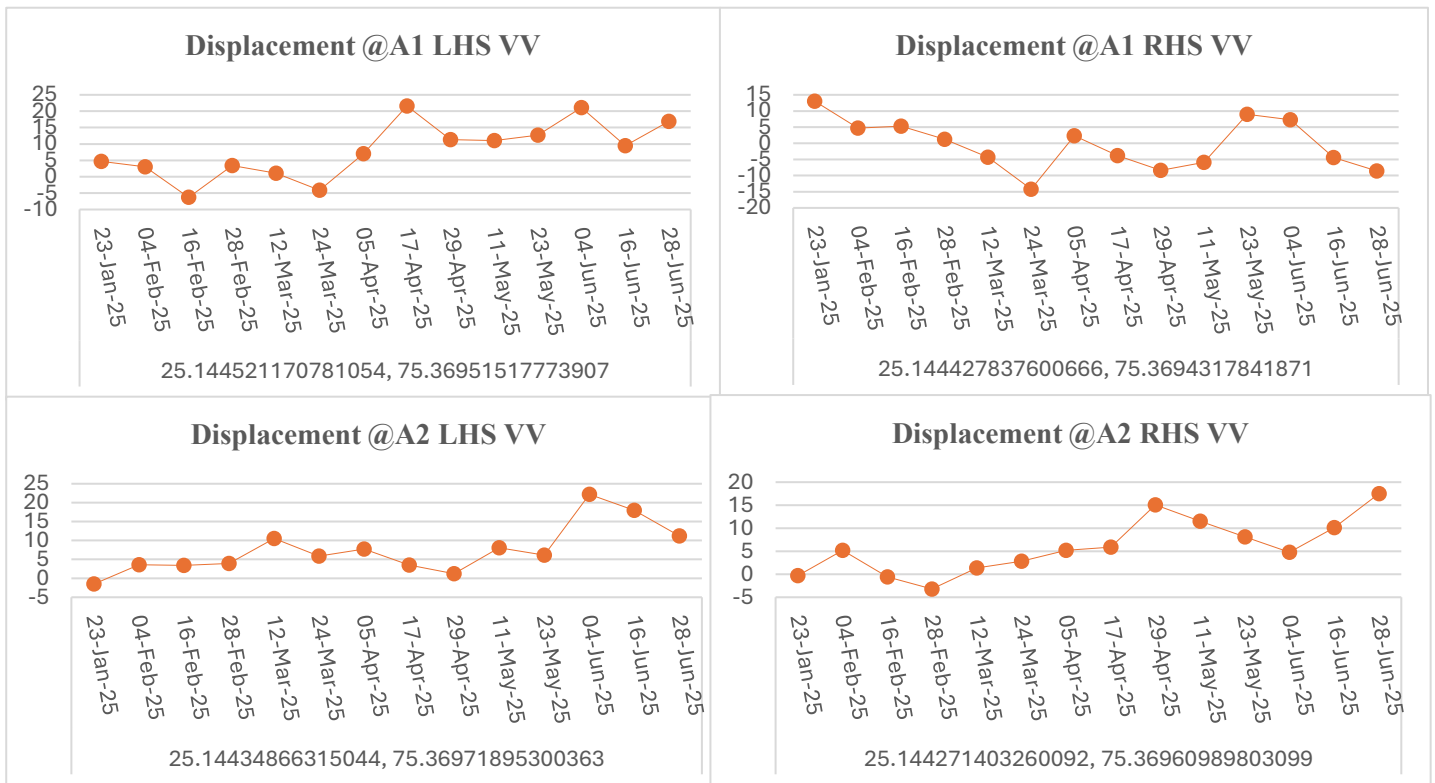
MNB – CH1006+100

Length – 42.51M

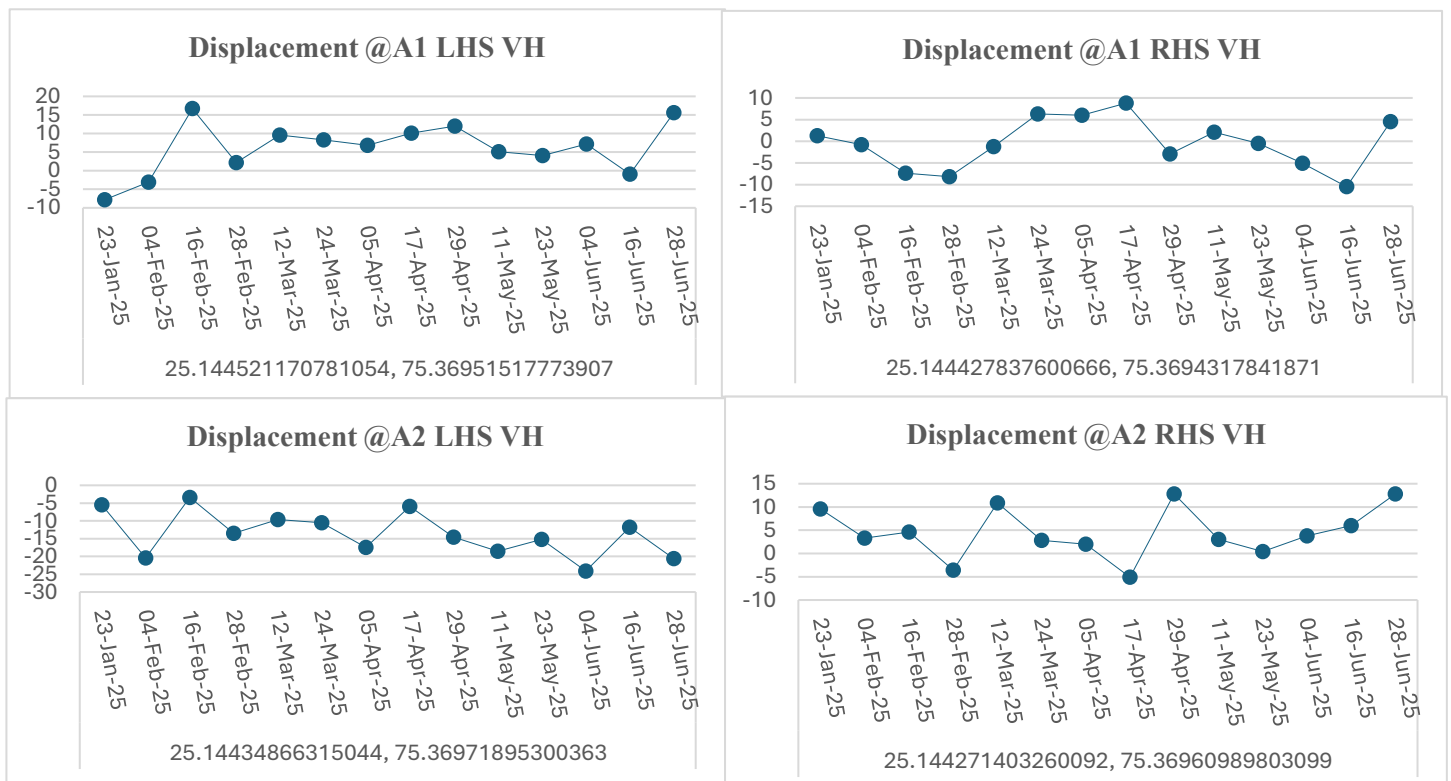


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

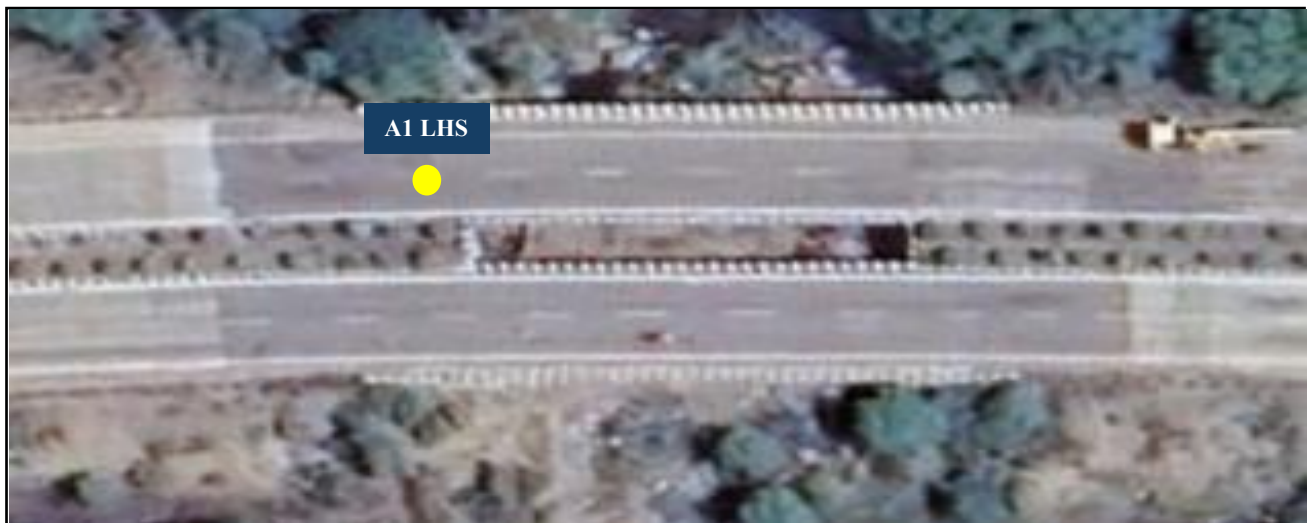
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



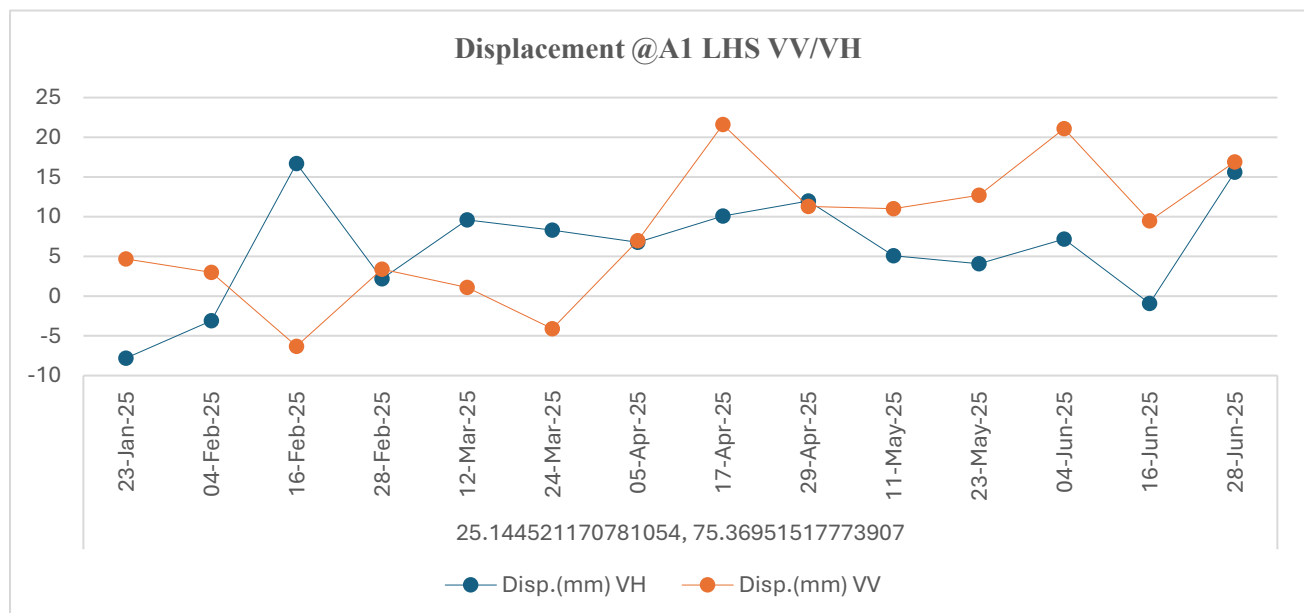
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.144521170781054, 75.36951517773907)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows moderate to high variability, alternating between settlement and uplift phases over the monitoring period (Jan–Jun 2025).
- **Horizontal displacement (VH)** Horizontal displacement exhibits high variability with sharp oscillations, spanning roughly from –8 mm to +16 mm. VH demonstrates frequent directional changes, suggesting unstable lateral behaviour with no sustained monotonic trend across the period

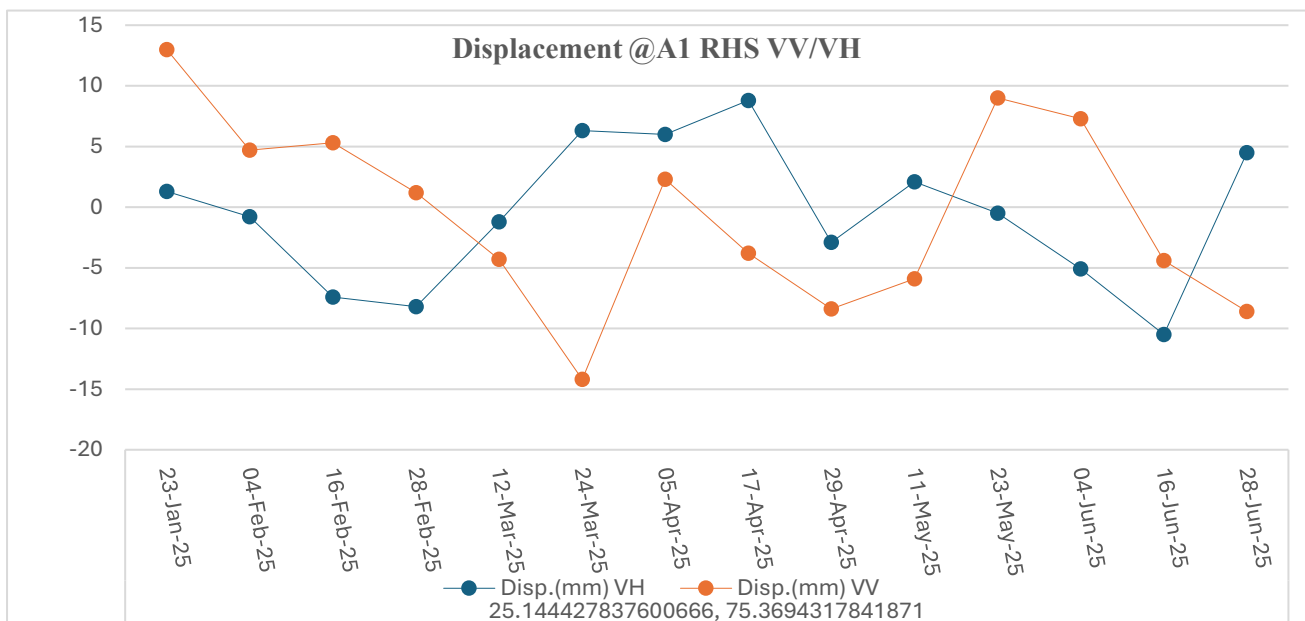
Critical Events & Anomalies:

- **January 2025:** VH begins at a negative minimum (–8 mm), while VV remains moderately positive (+4 mm), indicating initial lateral settlement without corresponding vertical subsidence.
- **May 2025:** VH decreases to +4 mm, while VV remains elevated (+13 mm), showing partial lateral relaxation under sustained uplift.
- **June 2025:** Both VH and VV increase concurrently (VH +15 mm, VV +16 mm), indicating a combined lateral and vertical displacement intensification toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.144427837600666, 75.3694317841871)



Key Observations:

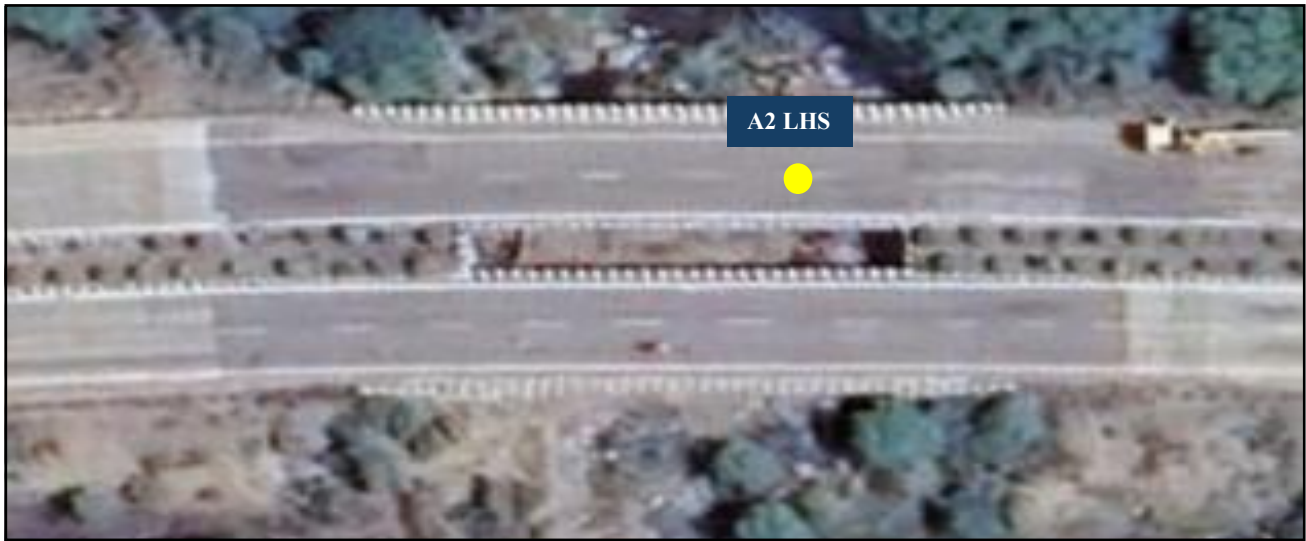
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate to high variability, oscillating between roughly -10 mm and +9 mm.
- **Vertical displacement (VV)** Vertical displacement shows high variability with frequent reversals, alternating between uplift and settlement throughout the monitoring period (Jan–Jun 2025).

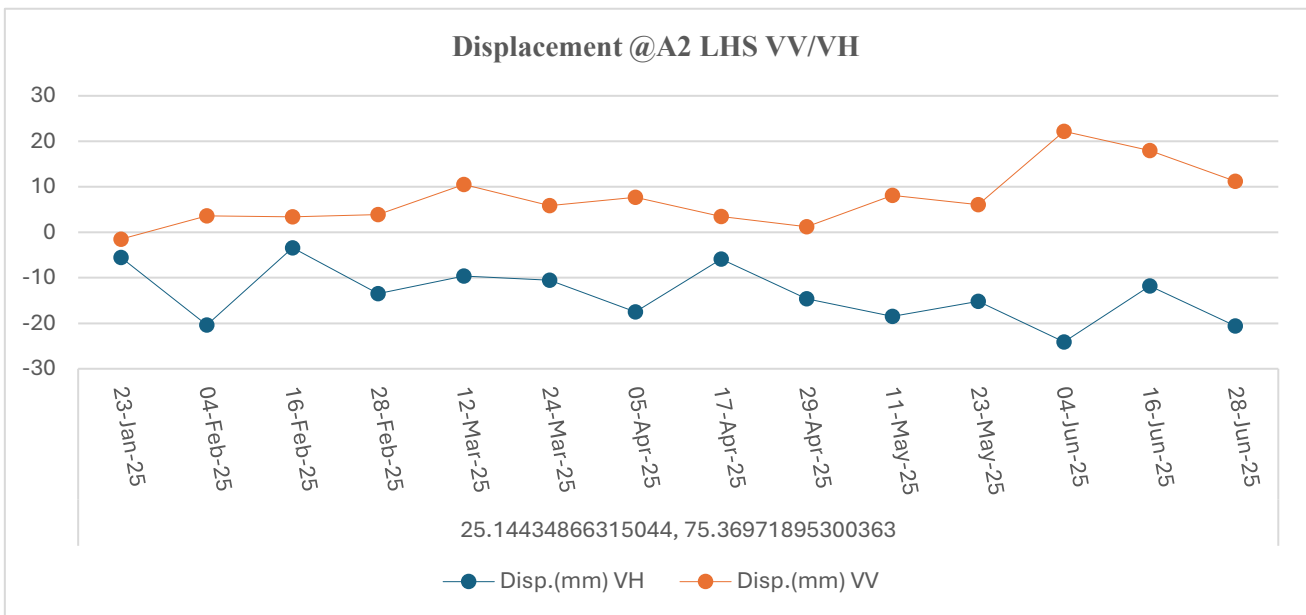
Critical Events & Anomalies:

- **January 2025:** VV starts at a strong positive peak (+13 mm), while VH remains near +1 mm, suggesting initial vertical uplift with minimal lateral movement.
- **April 2025:** VH drops again to -3 mm, while VV declines further to -8 mm, marking a combined settlement phase in both components.
- **June 2025:** VH recovers sharply to +4 mm, while VV remains negative (-8 mm), suggesting lateral rebound without vertical stabilization toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.14434866315044, 75.36971895300363)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV)** Vertical displacement is predominantly positive throughout the monitoring period (Jan–Jun 2025), indicating sustained vertical uplift.
- **Horizontal Displacement (VH)** Horizontal displacement remains consistently negative, indicating persistent lateral movement in a single dominant direction.

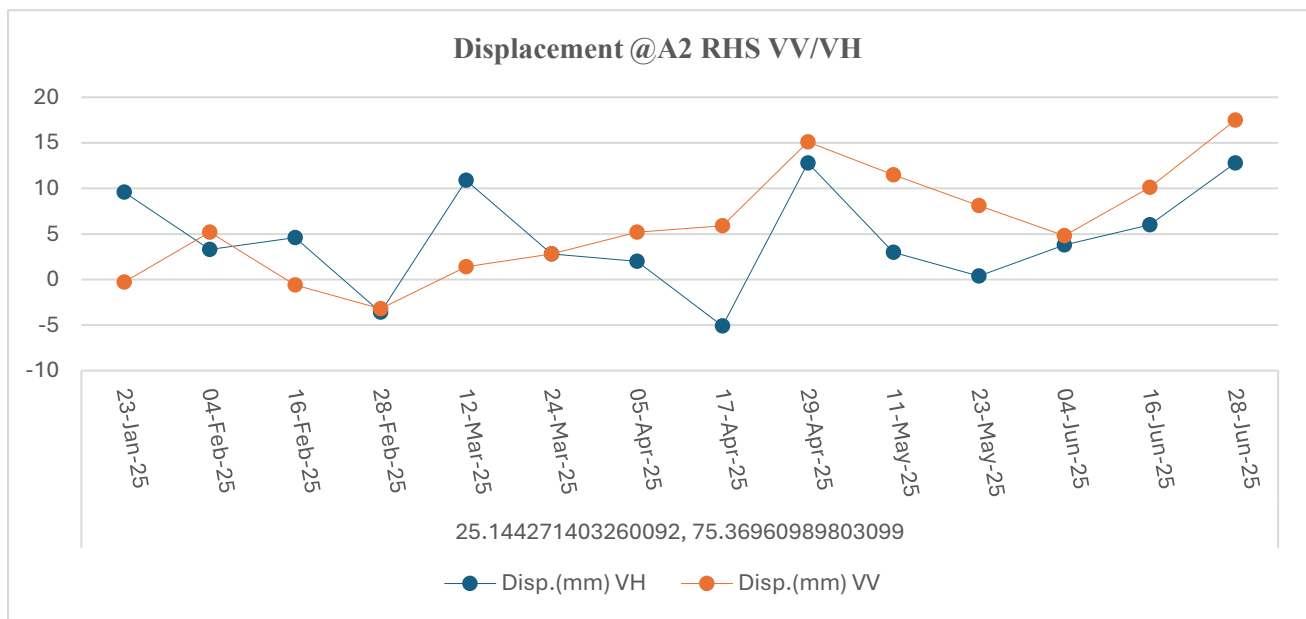
Critical Events & Anomalies:

- **January 2025:** VH begins near –8 mm, while VV is close to 0 mm, indicating initial lateral displacement with minimal vertical uplift.
- **April 2025:** VH improves briefly to –7 mm, while VV dips slightly to +2 mm, indicating a short-lived horizontal stabilization without loss of uplift.
- **June 2025:** VH partially recovers to –18 mm, while VV decreases to +11 mm, indicating post-peak adjustment rather than full stabilization.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.144271403260092, 75.36960989803099)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical displacement (VV)** Vertical displacement shows a clear upward trend over the monitoring period (Jan–Jun 2025), transitioning from near-zero or slight settlement in January to strong vertical uplift by late June.
- **Horizontal displacement (VH)** Horizontal displacement exhibits moderate variability with alternating uplift and settlement phases, ranging roughly between –5 mm and +13 mm.

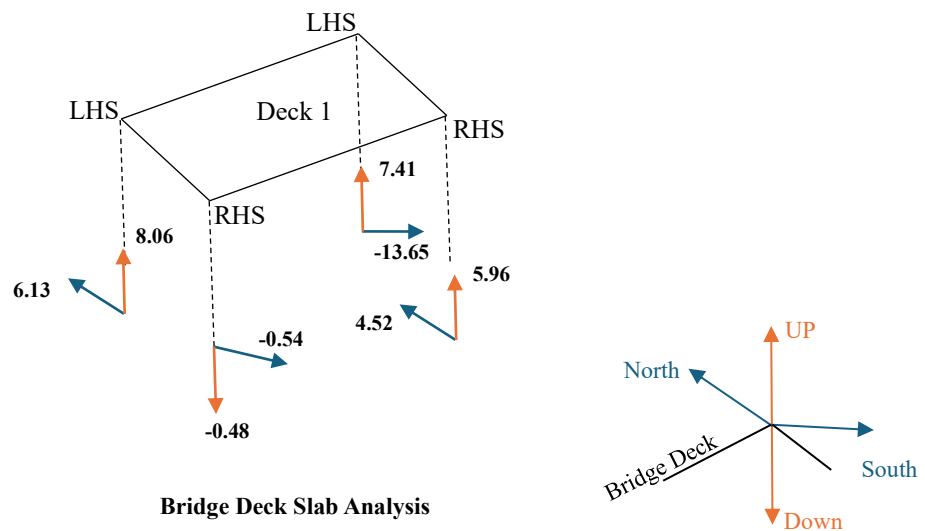
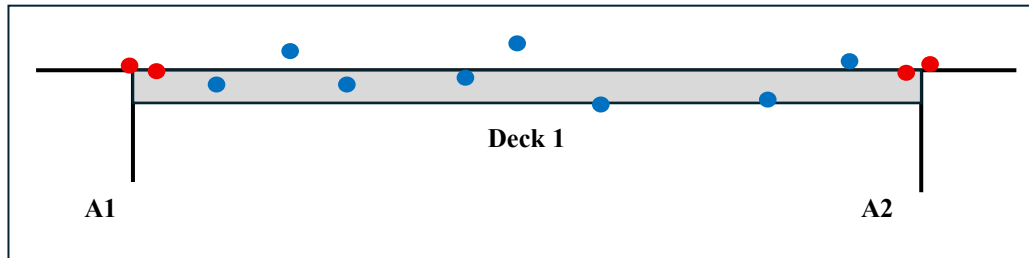
Critical Events & Anomalies:

- **January 2025:** VH starts at a positive value (+9 mm), while VV is near 0 mm, indicating initial lateral displacement with negligible vertical response.
- **April 2025:** VH drops to its most negative value (–5 mm), while VV continues increasing (+5 mm), highlighting a decoupled deformation phase dominated by vertical uplift.
- **June 2025:** Both VH and VV increase steadily, ending at VH +13 mm and VV +17 mm, indicating simultaneous vertical uplift and lateral displacement intensification toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- Vertical displacement across the bridge deck slab shows moderate differential movement, ranging from +8.06 mm uplift at the LHS upper point to -0.48 mm downward displacement at the RHS lower-central point. The RHS upper deck location records +7.41 mm uplift, while the RHS outer point shows +5.96 mm uplift, indicating a dominant upward response on the right-hand side of the deck.

Horizontal Displacement Behaviour:

- Horizontal displacements vary significantly in both magnitude and direction, ranging from -13.65 mm at the RHS upper point to +6.13 mm at the LHS outer point. The LHS region displays a northwest-directed horizontal movement, while the RHS and central point's show southeast to southward displacement components (-10.54 mm and -13.65 mm), confirming opposing lateral movement directions across the deck slab. This spatial contrast highlights a strong asymmetry in horizontal deformation behaviour.

CONCLUSION:

- The combined vertical and horizontal displacement patterns indicate a non-uniform and non-symmetric deformation response of the bridge deck slab. The RHS exhibits consistent uplift coupled with southward lateral movement, while the LHS shows smaller uplift but opposing horizontal displacement.